Inclusion Conjunctivitis (MONDO:0005808): A Comprehensive Disease Characteristics Report

Category: Infectious Disease **Causative agent:** *Chlamydia trachomatis* serovars D–K (obligate intracellular bacterium) **Report basis:** 14 confirmed findings across 37 reviewed papers; human clinical cohorts, case reports, reviews, and a guinea-pig animal model

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Summary

Inclusion conjunctivitis is an infectious follicular conjunctivitis caused by the obligate intracellular bacterium *Chlamydia trachomatis*, specifically the "oculogenital" serovars D–K — the same serovars that cause urogenital chlamydia. It is fundamentally an infectious, sexually-linked disease with no genetic etiology. The condition presents in two demographic settings: (1) **adult inclusion conjunctivitis (AIC)**, a sexually transmitted disease acquired by transfer of infected genital secretions to the eye (autoinoculation or sexual contact), peaking in young sexually active adults; and (2) **neonatal inclusion conjunctivitis (ophthalmia neonatorum)**, acquired by the newborn passing through an infected birth canal during vaginal delivery. This dual presentation reflects a single reservoir — genital *C. trachomatis* infection in the sexually active population.

Mechanistically, disease follows the hallmark chlamydial **biphasic developmental cycle**: infectious **elementary bodies (EBs)** are inoculated onto conjunctival epithelium, are endocytosed, and differentiate within a membrane-bound **inclusion** into replicative **reticulate bodies (RBs)** that divide by binary fission before re-converting to EBs and lysing the host cell to spread. The intracellular infection and released chlamydial antigens drive a **neutrophil- and lymphocyte-mediated immune response** producing the characteristic **subepithelial follicles (germinal centers)**, papillary hypertrophy, mucopurulent discharge, and preauricular adenopathy. In neonates, infection can extend beyond the conjunctiva to cause **chlamydial (interstitial) pneumonia**.

Inclusion conjunctivitis is clinically and epidemiologically **distinct from trachoma** (serovars A–C), which is a chronic, blinding, cicatricial keratoconjunctivitis of the upper tarsal conjunctiva driven by repeated childhood reinfection in endemic areas. By contrast, inclusion conjunctivitis is sporadic, generally non-blinding, and **curable with systemic antibiotics** — single-dose oral azithromycin 1 g or a doxycycline course in adults; erythromycin or azithromycin in neonates (with amoxicillin/macrolides for the maternal reservoir in pregnancy, since doxycycline is contraindicated). Diagnosis is best made by **nucleic acid amplification testing (NAAT/PCR)** of conjunctival specimens, which outperforms culture and Giemsa inclusion cytology. Prevention rests chiefly on **antenatal maternal screening and treatment**, not neonatal topical prophylaxis, which is inadequate against *Chlamydia*.

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1. Disease Information

Overview. Inclusion conjunctivitis is an infectious conjunctivitis caused by *Chlamydia trachomatis*. It is called "inclusion" conjunctivitis because of the pathognomonic intracytoplasmic **inclusion bodies** (chlamydial inclusions) formed in infected conjunctival epithelial cells. The disease sits at one pole of the oculogenital *C. trachomatis* disease spectrum, which — per a review of STI-associated eye infections — "*includes STIs, inclusion conjunctivitis in adults and neonatal ophthalmia*" (PMID: 35027140) (F013).

Key identifiers.

Resource	Identifier
Mondo	MONDO:0005808
Causative organism (NCBI Taxonomy)	*Chlamydia trachomatis* (txid813)
MeSH	Conjunctivitis, Inclusion; Chlamydia trachomatis
ICD-10	A74.0 (Chlamydial conjunctivitis); P39.1 (Neonatal conjunctivitis / ophthalmia neonatorum)
ICD-11	1C22 (Chlamydial infection); neonatal conjunctivitis under KA modules

Synonyms / alternative names: adult inclusion conjunctivitis (AIC); chlamydial conjunctivitis; oculogenital chlamydial conjunctivitis; paratrachoma; swimming-pool conjunctivitis (historical); neonatal chlamydial conjunctivitis / chlamydial ophthalmia neonatorum (neonatal form); inclusion blennorrhea (historical neonatal term).

Information source: The evidence base is drawn from **aggregated disease-level resources** — clinical cohorts, case series, reviews, and one animal model — rather than individual EHR records, although some cohorts (e.g., a Cosmos EHR study of neonatal conjunctivitis, PMID: 39749988) draw on de-identified patient data.

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2. Etiology

Primary cause — infectious. Inclusion conjunctivitis is caused entirely by infection with *C. trachomatis* serovars D–K. There is **no genetic etiology**; this is not a Mendelian or complex genetic disease. As stated for the adult form, "*Inclusion conjunctivitis caused by the serovars D-K of Chlamydia trachomatis also affects the sexually active population*" (PMID: 23692878) (F001).

Risk factors (environmental/behavioral):

- **Adults:** sexual activity and genital *C. trachomatis* infection; young adult age (peak 16–20 years); autoinoculation from infected genital secretions; sexual/orogenital contact. Adult chlamydial conjunctivitis "*is a complication of sexually transmitted genital infection by C trachomatis and it indicates a large reservoir of such infections*" (PMID: 7427689) (F014).
- **Neonates:** maternal genital chlamydial infection and **vaginal delivery**. In a 10-year cohort of 797 neonates with suspected ophthalmia neonatorum, vaginal delivery was among the strongest predictors of chlamydial infection (OR = 17.63, 95% CI 5.59–55.51, $p < 0.001$), alongside bloody discharge (OR = 20.99, 95% CI 6.09–72.29) (PMID: 41062949) (F002).

Genetic risk / protective factors: Not applicable — no causal or susceptibility loci; no protective alleles. Host genetic modifiers of chlamydial immunopathology are relevant to blinding trachoma but not established for inclusion conjunctivitis.

Gene–environment interactions: Not applicable for this infectious disease.

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3. Phenotypes

Inclusion conjunctivitis presents as a **subacute-to-chronic follicular conjunctivitis**. In a clinical/molecular AIC cohort, it presented with "*variable hyperaemia, follicular reaction, papillary hypertrophy and mucopurulent discharge*" (PMID: 42350099) (F004).

Phenotype	Type	HPO suggestion	Characteristics / Frequency
Follicular conjunctivitis (subepithelial follicles)	Clinical sign	HP:0200019 (related)	**Most common sign**; earliest to regress after treatment (PMID: 24047438) (F004)
Conjunctival hyperemia / redness	Clinical sign	HP:0025421 (Conjunctival hyperemia)	Variable; common
Mucopurulent discharge	Symptom/ sign	HP:0000509 (related)	Common
Papillary hypertrophy	Clinical sign	—	Common
Preauricular lymphadenopathy	Clinical sign	HP:0410019 (regional lymphadenopathy)	Characteristic of follicular chlamydial conjunctivitis
Pseudoptosis	Clinical sign	HP:0000508 (Ptosis, related)	Conspicuous in some adult cases (PMID: 3838901) (F014)
Foreign-body sensation / irritation	Symptom	—	Common
Chronic, therapy-resistant course	Course feature	—	Frequently misdiagnosed (PMID: 28443807) (F004)

Onset & severity: Adult form typically presents in young adults; neonatal form within days to weeks of birth. Severity is mild-to-moderate and generally non-blinding, but the course can be **prolonged if unrecognized/untreated** ("*a prolonged course is expected if unrecognized/ untreated*," F014). Laterality is characteristically **unilateral** in adults: "*In patients with proved chlamydial conjunctivitis unilateral symptoms were the rule*" (PMID: 3838901) (F014).

Quality of life: Chronic irritation, discharge, and diagnostic delay (with unnecessary workups for orbital or thyroid eye disease, PMID: 35623592) impose morbidity, but the disease is not sight-threatening in the great majority of cases and resolves with appropriate therapy.

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4. Genetic/Molecular Information

Not applicable — this is an infectious, non-genetic disease. There are no human causal genes, pathogenic variants, modifier genes, epigenetic lesions, or chromosomal abnormalities.

The relevant molecular biology is that of the **pathogen**: the *C. trachomatis* **ompA** gene (encoding the major outer membrane protein, MOMP) defines serovars/genotypes D–K and is used for molecular typing. An AIC cohort "*provides a clinical and molecular profile of AIC and highlights variability across ompA serotypes*" (PMID: 42350099) (F001). In the pathogen, azithromycin resistance can arise from single point mutations in **23S rRNA (A2058/A2059)**, demonstrated in the related *C. caviae* (F011, PMID: 20065052).

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5. Environmental Information

Infectious agent (organism): *Chlamydia trachomatis*, an obligate intracellular Gram-negative bacterium. Ocular disease is divided by serovar: serovars **A, B, C** cause trachoma; serovars **D–K** cause inclusion conjunctivitis and urogenital infection (F006, F001).

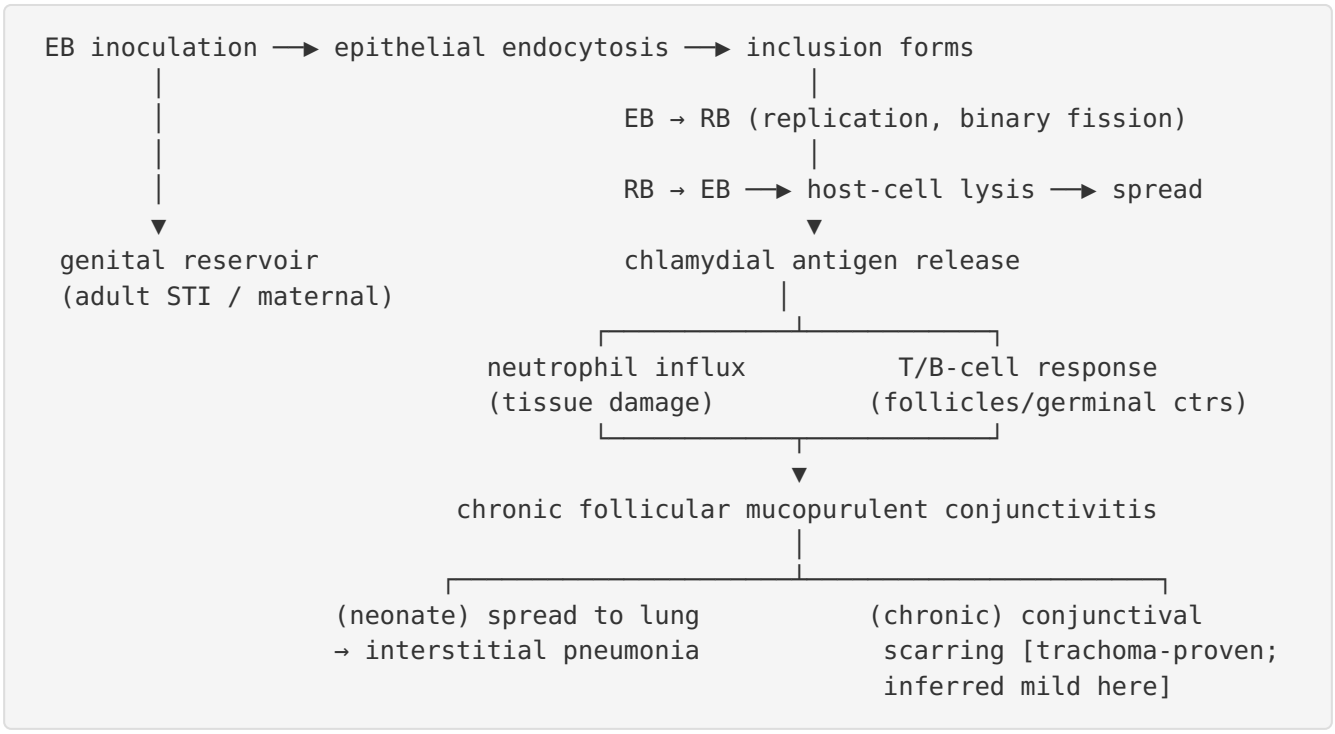
Environmental/behavioral factors: sexual transmission of genital chlamydia; transfer of infected secretions to the eye; passage through an infected birth canal. Historically, contaminated swimming-pool water was implicated ("swimming-pool conjunctivitis"). No toxin, radiation, pollution, occupational-exposure, or lifestyle (smoking/diet/alcohol) factor is causally established beyond sexual behavior as the route to genital infection.

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6. Mechanism / Pathophysiology

Ordered causal chain (initiating infection → clinical manifestation)

1. **Inoculation.** Infectious **elementary bodies (EBs)** are deposited on the conjunctival epithelium — from infected genital secretions (adult autoinoculation/sexual contact) or from the birth canal during vaginal delivery (neonate). *(Demonstrated route; F001, F002.)*
2. **Attachment and entry** — EBs attach to and are endocytosed by conjunctival epithelial cells, entering a membrane-bound compartment, the **chlamydial inclusion**. *"Chlamydia are obligate intracellular bacteria that reside within a membrane-bound compartment called the chlamydial inclusion inside a eukaryotic host cell"* (PMID: 35703434) (F009).
3. **EB → RB differentiation and replication** — inside the inclusion, EBs convert to replicative **reticulate bodies (RBs)** that divide by binary fission. *"Chlamydia undergo a biphasic developmental cycle alternating between the infectious elementary body (EB) and the replicative reticulate body (RB)"* (PMID: 36385643) (F009).
4. **RB → EB re-conversion and release** — *"RBs replicate by binary fission within an infected host cell, but there is a delay before RBs convert into EBs for spread to a new host cell"* (PMID: 33594486); host-cell lysis/extrusion releases progeny EBs that infect neighboring epithelium (F009).
5. **Innate immune activation** — infection and released chlamydial antigens trigger innate responses; **neutrophils** infiltrate and drive tissue pathology (established causally in the animal model: neutrophil depletion "*dramatically decreased ocular pathology, both clinically and histologically*," PMID: 21402767) (F011).
6. **Adaptive immune response** — lymphocytic (CD4+/CD8+ T-cell) infiltration and B-cell responses form **subepithelial follicles (germinal centers)** and papillary hypertrophy (F009, F011).
7. **Clinical disease** — chronic follicular mucopurulent conjunctivitis, hyperemia, preauricular adenopathy (F004).
8. **Branch (neonate):** the same organism can spread to the respiratory tract, causing **chlamydial interstitial pneumonia** (F007, F013).
9. **Branch (chronic/repeated infection):** prolonged immunopathology can cause conjunctival scarring — **demonstrated for trachoma** (serovars A–C, PMID: 26424969); *inferred to be milder and usually non-cicatricial* in inclusion conjunctivitis (F006, F009).



Molecular pathways / cellular processes (GO suggestions): intracellular bacterial infection/entry into host (GO:0044409); host innate immune response (GO:0045087); inflammatory response (GO:0006954); adaptive immune response / germinal center formation (GO:0002250, GO:0002467). **Cell types (CL):** conjunctival epithelial cell (CL:0000066 epithelial cell), neutrophil (CL:0000775), CD4+ T cell (CL:0000624), CD8+ T cell (CL:0000625), B cell/plasma cell. In the animal model, neutrophil depletion also raised *C. caviae*-specific tear/serum IgA and altered IL-5, TGF-β, and CCL5/RANTES, showing neutrophils shape both innate damage and the adaptive response (F011).

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7. Anatomical Structures Affected

- **Primary organ/tissue:** the **conjunctiva** (UBERON:0001811), specifically the **conjunctival epithelium** (epithelial tissue). Palpebral/tarsal and bulbar conjunctiva are involved; follicles predominate in the inferior fornix/tarsal conjunctiva.
- **Associated structures:** cornea (UBERON:0000964) may show punctate epithelial keratitis or micropannus in some cases; regional **preauricular lymph nodes** enlarge.
- **Secondary organ involvement (neonate): lung** (UBERON:0002048) — chlamydial interstitial pneumonia (F007).

- **Subcellular (GO Cellular Component):** the pathogen occupies the **chlamydial inclusion**, a modified host membrane-bound vacuole in the cytoplasm; host cytoplasm and endomembrane system are co-opted (F009).
- **Body systems:** ocular/visual system; reproductive/genitourinary system (reservoir); respiratory system (neonatal complication).
- **Lateralization:** characteristically **unilateral** in adults (F014); neonatal disease is often bilateral.

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8. Temporal Development

- **Onset:** Adult form — subacute to insidious, in sexually active young adults (peak 16–20 years, F014). Neonatal form — within the first days to ~2 weeks of life following delivery.
- **Onset pattern:** subacute/chronic; frequently indolent and therapy-resistant if treated only with topical agents (F004).
- **Course:** untreated disease can be **prolonged/chronic** (F014). Follicles are the earliest sign to regress with successful systemic treatment, though **complete relief may take ~1 month** (PMID: 24047438) (F003, F004).
- **Resolution:** treatment-induced remission with systemic antibiotics; prompt therapy "*can eradicate the microorganism at least in uncomplicated infections, preventing the sequelae of the disease*" (PMID: 2247702) (F013).
- **Critical period:** the perinatal window (maternal screening/treatment before delivery) is the key opportunity to prevent neonatal disease (F005).

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9. Inheritance and Population

Inheritance: Not applicable — infectious, non-heritable.

Epidemiology. The disease tracks the burden of genital chlamydia. A review estimates global *C. trachomatis* prevalence at **~2.8% in men and ~3.8% in women** (PMID: 35027140) (F013).

Setting	Finding	Source
Eye-clinic conjunctivitis patients	Chlamydia in 28/983 (~2.8%); peak >9% at ages 16–20	PMID: 3838901 (F14)
Neonatal ophthalmia (Singapore, n=797)	*C. trachomatis* leading cause: 140 (17.5%)	PMID: 41062949 (F02)
Symptomatic neonatal conjunctivitis (Hungary, n=76)	*C. trachomatis* in 30 (39.5%)	PMID: 28693667 (F12)
Perinatal transmission (preterm cohort)	maternal carriage ~10%; mother-to-infant transmission 75%	PMID: 22568356 (F07)

Serovar/genotype distribution: oculogenital serovars **E and G predominate**, mirroring urogenital chlamydia. In the Hungarian neonatal series: genotype **E 48%** (14/29), **G 24%** (7/29), J 10%, K and F 6.9% each, H 3.4% (PMID: 28693667) (F12).

Demographics: adults — young, sexually active; neonates — infants of infected mothers, especially after vaginal delivery. **Age peak 16–20 years** in adults; not seen in ages 1–15 (F14). Geographic distribution is worldwide and sporadic (non-endemic), unlike trachoma.

10. Diagnostics

NAAT/PCR is the most sensitive test. In 241 neonates, *C. trachomatis* was detected by PCR in 16.6%, culture in 12.9%, and Giemsa staining in only 7.5%; PCR showed "*High sensitivity (100%) and specificity (95.7%)... as compared to culture*" (PMID: 22369307) (F10).

Modality	Role	Relative yield
NAAT / PCR (e.g., transcription-mediated amplification, Aptima Combo 2)	**Test of choice**; confirms atypical cases	Highest (100% sens vs culture)
Cell culture	Historical gold standard	Intermediate (12.9%)
Giemsa cytology (intracytoplasmic inclusions)	Rapid, low-cost	Lowest (7.5%)
Conjunctival cytology + NAAT + culture combined	Comprehensive AIC workup	—

Contemporary AIC cohorts combine cytology, NAAT, and culture (PMID: 42350099); atypical adult presentations were confirmed by TMA (PMID: 35623592) (F10).

Clinical clue / differential: Persistent, therapy-resistant follicular conjunctivitis should prompt chlamydial testing — "*when a patient presents with persistent, therapy-resistant conjunctivitis, particularly if conjunctival follicles are present, chlamydial conjunctivitis should be considered*" (PMID: 28443807) (F04). Differential diagnosis includes viral (adenoviral) follicular conjunctivitis, other bacterial conjunctivitis, allergic conjunctivitis, keratoconjunctivitis sicca, and — critically — **trachoma** in endemic settings. Adult diagnosis should trigger full genital/serological STI investigation (F14).

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11. Outcome / Prognosis

Excellent with prompt antibiotics. Prompt treatment eradicates the organism in uncomplicated infection and prevents sequelae (PMID: 2247702) (F13). Unlike blinding trachoma, inclusion conjunctivitis is **generally non-blinding** (F06).

- **Mortality:** essentially nil for the ocular disease itself.
- **Morbidity:** chronic ocular irritation, misdiagnosis/diagnostic delay; complete symptom relief may take ~1 month even after appropriate therapy (F03).
- **Neonatal complications:** spread of infection to cause **chlamydial pneumonia** (all infected infants in one preterm cohort developed respiratory disorders/pneumonia, some requiring nCPAP or ventilation, PMID: 22568356); maternal chlamydia is associated with low birth weight and preterm birth (F07, PMID: 27144177, PMID: 27164464).
- **Prognostic factors:** promptness of systemic (not merely topical) therapy; recognition of the diagnosis; treatment of the genital reservoir and partners.

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12. Treatment

Systemic antibiotics are required — chlamydial conjunctivitis "*warrant[s] systemic antibiotics due to their potential for severe complications*," unlike routine bacterial conjunctivitis (PMID: 37977204) (F03).

Population	Regimen	Evidence
Adults	**Single-dose oral azithromycin 1 g** (NCIT: Azithromycin) — equivalent to prolonged azithromycin and 21-day doxycycline for retreatment rate	RCT, n=83: "*All treatment groups provided statistically equivalent results of retreatment rate*" (PMID: 24047438) (F03)
Adults (alternative)	Doxycycline course; oral azithromycin 1 g resolved atypical cases	PMID: 35623592 (F03)
Neonates	Systemic erythromycin or azithromycin	F05, F13
Pregnant women (reservoir)	**Amoxicillin 7-day course** or erythromycin; **doxycycline & fluoroquinolones contraindicated**	"*Amoxicillin (a 7-day course) is the antibiotic of choice for pregnant women...*"; "*Doxycycline and fluoroquinolones are contraindicated during pregnancy*" (PMID: 22216551) (F08)

Adjuncts / strategy: partner treatment and full STI evaluation in adults; treat the maternal genital infection to protect the neonate (F08, F14). **NCIT suggestions:** Azithromycin (C551), Doxycycline (C459), Erythromycin (C576), Amoxicillin (C398). **Pharmacogenomics / resistance:** azithromycin resistance can arise from single 23S rRNA A2058/A2059 mutations (shown in *C. caviae*; attenuates virulence) (F11).

13. Prevention

Antenatal maternal screening and treatment is the most effective prevention. "*Prenatal screening and treatment of pregnant women is the most effective way to prevent ophthalmia neonatorum*" (PMID: 36691840) (F05).

- **Neonatal topical prophylaxis is inadequate against chlamydia.** "*Administration of erythromycin ophthalmic ointment for the prevention of neonatal conjunctivitis is not literature-supported*" (PMID: 36691840); a Botswana series found most infants who developed chlamydial ophthalmia had received ocular tetracycline prophylaxis — "*current prophylaxis and treatment modalities... are inadequate*" (PMID: 37338101) (F05).

- **Primary prevention:** safe-sex practices; STI screening in sexually active adults.
- **Secondary prevention:** early detection/treatment of genital chlamydia; partner notification and treatment.
- **Public health:** treating the "large reservoir" of genital infection reduces both adult and neonatal ocular disease (F14).
- **Immunization:** no licensed chlamydia vaccine (research ongoing).

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14. Other Species / Natural Disease

- **Guinea pig (*Cavia porcellus):* natural host of** Chlamydia caviae**, which causes *guinea-pig inclusion conjunctivitis (GPIC)** — "*Chlamydia caviae, which is a natural pathogen of guinea pigs*" (PMID: 20065052) (F11). This is a naturally occurring analog of human inclusion conjunctivitis and the principal animal model.
- **Cross-species relevance:** the EB→RB developmental cycle and inclusion biology are conserved across the *Chlamydiaceae*, underpinning the model's validity.
- **Zoonotic note:** *C. caviae* has occasionally caused human respiratory disease, but ocular inclusion conjunctivitis in humans is caused by *C. trachomatis*, not *C. caviae*.

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15. Model Organisms

Guinea pig *C. caviae* ocular infection (GPIC) is the principal animal model of inclusion conjunctivitis. It recapitulates ocular follicular immunopathology and enables mechanistic and antibiotic studies:

- **Neutrophil role:** neutrophil depletion "*dramatically decreased ocular pathology, both clinically and histologically*," and shifted adaptive immunity (higher *C. caviae*-specific IgA in tears/serum; fewer conjunctival CD4+/CD8+ T cells; altered IL-5, TGF-β, CCL5/RANTES) — establishing neutrophils as drivers of tissue damage (PMID: 21402767) (F11).
- **Antibiotic pharmacology:** spontaneous azithromycin resistance maps to single 23S rRNA mutations (A2058/A2059) and attenuates virulence in vivo (PMID: 20065052) (F11).

Cell-based systems: *C. trachomatis* is routinely studied in cultured epithelial cell lines for the developmental cycle, inclusion kinetics, and drug inhibition (PMID: 35703434, PMID: 33051378).

Limitations: the guinea-pig model uses a different species (*C. caviae*, not *C. trachomatis*) and does not fully replicate human sexual/perinatal transmission dynamics; cell culture omits the intact ocular immune microenvironment.

Mechanistic Model / Interpretation

Inclusion conjunctivitis is best understood as **one clinical face of a single genital-chlamydia reservoir**. The identical serovars D–K that colonize the urogenital tract reach the eye by two routes — adult autoinoculation/sexual contact, and neonatal passage through the birth canal — and there execute the universal chlamydial developmental program. The **EB→RB→EB cycle inside conjunctival epithelial inclusions** is the mechanistic engine; the **host neutrophil-then-lymphocyte response** converts intracellular infection into the visible follicular disease. This explains the key clinical facts: the chronic, therapy-resistant nature (intracellular persistence), the follicular morphology (germinal-center immunopathology), the requirement for **systemic** antibiotics (topical agents cannot reliably clear an intracellular, genitally-seeded infection), and the neonatal pneumonia branch (same organism, different mucosa).

The **trachoma vs. inclusion conjunctivitis** distinction is central and is defined by serovar, site, transmission, and outcome:

Feature	Inclusion conjunctivitis	Trachoma
Serovars	**D–K** (oculogenital)	**A, B, C**
Transmission	Sexual/perinatal, sporadic	Eye-to-eye, endemic, repeated childhood reinfection
Primary site	Conjunctiva (fornix/tarsal)	Upper tarsal conjunctiva
Course	Acute–chronic, usually **non-blinding**	Chronic cicatricial, **blinding**
Prognosis	Curable with systemic antibiotics	Leading infectious cause of blindness

Source: PMID: 26424969, PMID: 35027140 (F06, F13).

Evidence Base

PMID	Contribution	Finding
23692878	Serovars D–K cause adult inclusion conjunctivitis in sexually active population	F01
42350099	Clinical/molecular AIC profile; ompA serotype variability; presentation features	F01, F04, F10
41062949	*C. trachomatis* leading cause of ophthalmia neonatorum (17.5%); vaginal delivery OR 17.63	F02
24047438	RCT: single-dose azithromycin equivalent to longer regimens; follicles most common sign	F03, F04
37977204	Chlamydial conjunctivitis requires systemic antibiotics	F03
35623592	Atypical adult cases; oral azithromycin 1 g resolution; TMA diagnosis	F03, F04, F10
28443807	Persistent therapy-resistant follicular conjunctivitis → suspect chlamydia	F04
36691840	Prenatal screening/treatment most effective; topical prophylaxis not supported	F05
37338101	Field evidence topical prophylaxis fails against chlamydia	F05
26424969	Trachoma (A,B,C) distinct blinding disease	F06
22568356	75% perinatal transmission; all infected infants developed pneumonia	F07
22216551	Amoxicillin in pregnancy; doxycycline/fluoroquinolones contraindicated	F08
35703434	Obligate intracellular niche; chlamydial inclusion	F09
36385643	Biphasic EB–RB developmental cycle	F09
33594486	RB binary fission and RB→EB conversion for spread	F09
22369307	PCR 100% sens/95.7% spec vs culture; PCR>culture>Giemsa	F10
21402767	Guinea pig GPIC model; neutrophils drive pathology	F11
20065052	*C. caviae* natural guinea-pig pathogen; 23S rRNA azithromycin resistance	F11
28693667	Genotype E (48%) and G (24%) predominance; 39.5% positivity	F12
35027140		F13

PMID	Contribution	Finding
	Oculogenital spectrum incl. inclusion conjunctivitis; prevalence ~2.8%/ 3.8%	
2247702	Vertical transmission → inclusion conjunctivitis/interstitial pneumonia; prompt antibiotics eradicate	F13
3838901	Age peak 16–20 y (>9%); unilateral presentation	F14
7427689	Adult/neonatal chlamydial conjunctivitis marks large genital reservoir	F14

Limitations and Knowledge Gaps

- **No omics/molecular-profiling data specific to human ocular inclusion conjunctivitis** (transcriptomics, proteomics, metabolomics, single-cell) were identified; mechanistic immunology relies on the guinea-pig *C. caviae* model and general chlamydial cell biology.
- **Quantitative human immunopathology** (cytokine profiles, follicle histology grading) in inclusion conjunctivitis specifically is sparse; neutrophil causal data come from the animal model.
- **Frequency percentages for individual phenotypes** in adults are not precisely quantified across large cohorts; several estimates are qualitative.
- **Precise incidence/prevalence of the ocular disease** (as opposed to genital chlamydia) is not well defined; figures are inferred from conjunctivitis cohorts and STI prevalence.
- **Scarring risk in inclusion conjunctivitis** is inferred (by analogy to trachoma) to be mild/ uncommon rather than directly quantified.
- Some cited abstracts (e.g., PMID: 42350099, PMID: 31753423) had limited or unavailable full text.

Proposed Follow-up Experiments / Actions

1. **Human ocular immunoprofiling:** single-cell RNA-seq / cytokine profiling of conjunctival specimens from confirmed AIC patients to directly test the neutrophil→lymphocyte follicle model in humans (currently only guinea-pig evidence).

2. **Serovar–phenotype correlation:** larger ompA-genotyped AIC cohorts to test whether serovars E/G differ in severity, chronicity, or treatment response.
3. **Antimicrobial resistance surveillance:** monitor 23S rRNA (A2058/A2059) and other resistance markers in ocular *C. trachomatis* isolates given single-dose azithromycin use.
4. **Prevention policy evidence:** prospective evaluation of universal antenatal chlamydia screening vs. neonatal topical prophylaxis for preventing neonatal ocular and pulmonary disease.
5. **Diagnostic pathways:** embed reflex NAAT for persistent/therapy-resistant follicular conjunctivitis to reduce the well-documented diagnostic delay and unnecessary orbital/thyroid workups.
6. **Epidemiologic quantification:** registry-based estimation of true incidence/prevalence of ocular (vs. genital) chlamydial disease, and of neonatal pneumonia progression rates.

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*Report compiled from 14 confirmed findings and 37 reviewed papers. Evidence types: human clinical cohorts and case series, narrative reviews, and a guinea-pig (*C. caviae*) animal model plus in vitro chlamydial cell biology. Inclusion conjunctivitis is an infectious, non-genetic disease; genetic/genomic sections are annotated "not applicable" where appropriate.*
